

ON AN INVERSE BOUNDARY PROBLEM IN ELASTODYNAMICS

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ABSTRACT. We study the simultaneous identification of the Lamé parameters λ and μ in three-dimensional static isotropic elasticity. For smooth coefficients, we establish uniqueness from the full Dirichlet-to-Neumann map when μ is axisymmetric, admits a multiplicative or additive separation on a product domain, or is quasianalytic in one fixed direction. These results require neither smallness of $\nabla\mu$ nor real analyticity of the coefficients, and impose no corresponding structural condition on λ . In the axisymmetric and separated cases, partial boundary data suffice to determine μ .

1. INTRODUCTION

1.1. Problem formulation and background. Let $\Omega \subset \mathbb{R}^3$ be a known bounded connected Lipschitz domain. For a displacement $u : \Omega \rightarrow \mathbb{R}^3$, write

$$\varepsilon(u) = \frac{1}{2}(\nabla u + (\nabla u)^T), \quad \sigma_{\lambda,\mu}(u) = \lambda(\operatorname{div} u)I + 2\mu\varepsilon(u),$$

where λ and μ are the *Lamé parameters* and μ is the *shear modulus*. We assume that the coefficients are real valued and bounded, and satisfy the uniform strong convexity condition

$$\mu \geq c_*, \quad 3\lambda + 2\mu \geq c_* \quad \text{almost everywhere in } \Omega \quad (1.1)$$

for some $c_* > 0$. The *elastic Dirichlet problem*

$$\mathcal{L}_{\lambda,\mu}u := \operatorname{div} \sigma_{\lambda,\mu}(u) = 0 \quad \text{in } \Omega, \quad u|_{\partial\Omega} = f \quad (1.2)$$

has a unique weak solution $u_f \in H^1(\Omega)^3$ for every $f \in H^{1/2}(\partial\Omega)^3$ [3]. Its Dirichlet-to-Neumann map is

$$\Lambda_{\lambda,\mu}^\Omega : H^{1/2}(\partial\Omega)^3 \longrightarrow H^{-1/2}(\partial\Omega)^3, \quad \Lambda_{\lambda,\mu}^\Omega f = \sigma_{\lambda,\mu}(u_f)\nu|_{\partial\Omega}, \quad (1.3)$$

where ν is the outward unit normal and the traction is understood in the weak sense. We consider the inverse problem of recovering both Lamé parameters from knowledge of $\Lambda_{\lambda,\mu}^\Omega$. We are mainly concerned with the unique identifiability issue, namely, sufficient conditions under which

$$\Lambda_{\lambda_1,\mu_1}^\Omega = \Lambda_{\lambda_2,\mu_2}^\Omega \implies (\lambda_1, \mu_1) = (\lambda_2, \mu_2) \quad \text{in } \Omega. \quad (1.4)$$

We also consider *partial boundary data*, for which only a nonempty relatively open subset Γ of the smooth part of $\partial\Omega$ is accessible. Displacements are prescribed on Γ and set to zero on $\partial\Omega \setminus \Gamma$, and the resulting traction is measured only on Γ . We denote the corresponding localized Dirichlet-to-Neumann map by $\Lambda_{\lambda,\mu}^\Gamma$; its precise definition is given in Section 2.1.

Two established approaches to this problem are relevant to the present work. Perturbative uniqueness is known when $\nabla\mu$ is sufficiently small in a sufficiently high C^k norm, under the corresponding regularity and a priori bounds; see the results and corrected theorem

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in [5, 6, 11]. With partial boundary data, both parameters can be recovered when μ is constant and the boundary values of λ are known on the inaccessible part [8]. Another approach combines boundary determination with real analyticity. Under suitable smoothness hypotheses, boundary measurements determine the Taylor coefficients of the Lamé parameters [9, 10]. If both parameters are real analytic in a neighborhood of the closed connected domain, these boundary values and derivatives determine them throughout the domain by analytic continuation [13]. These results motivate the search for uniqueness beyond nearly constant shear moduli and real-analytic Lamé parameters.

In this paper, we show how the structure and regularity of the shear modulus lead to unique identifiability in isotropic elasticity. Symmetry and separation of variables yield uniqueness for smooth coefficients, while quasianalyticity provides a complementary route by propagating boundary information into the interior. Our main results are as follows.

- (i) **Axisymmetric shear moduli.** We establish unique identifiability of axisymmetric shear moduli on domains of revolution from full torsional boundary data in the bounded measurable setting, and from partial torsional data in the smooth setting.
- (ii) **Separation of variables.** We prove that partial boundary data uniquely determine a smooth shear modulus admitting either a multiplicative or an additive separation on a product domain, even when both factors or both summands are unknown.
- (iii) **Directional quasianalyticity.** We establish unique identifiability of smooth shear moduli that are quasianalytic in one fixed direction from full boundary data, without requiring the domain to be convex.

In each of the smooth settings above, the full elastic Dirichlet-to-Neumann map uniquely determines both Lamé parameters, with the structural or quasianalytic condition imposed only on the shear modulus.

1.2. Main results. We state our main results for axisymmetric, separated, and directionally quasianalytic shear moduli.

1.2.1. Axisymmetric shear moduli. Let $D \subseteq \{(r, z) \in \mathbb{R}^2 : r > 0\}$ be a bounded connected domain with C^∞ boundary, and define its solid of revolution by

$$\Omega_D = \{(r \cos \theta, r \sin \theta, z) : (r, z) \in D, 0 \leq \theta < 2\pi\}.$$

The assumption on D means that $0 < r_- \leq r \leq r_+ < \infty$ on $\overline{D}$. Write $e_\theta = (-\sin \theta, \cos \theta, 0)$. For $f \in H^{1/2}(\partial D)$, the torsional boundary displacement is

$$Tf = rf(r, z)e_\theta.$$

Its boundedness as a trace map is proved in Proposition 3.2. If $\tilde{\Gamma} \subset \partial D$ is a nonempty relatively open arc, let $\Gamma \subset \partial\Omega_D$ be the surface obtained by rotating that arc. Define

$$\mathcal{T}_{\lambda, \mu}(f, g) = \langle \Lambda_{\lambda, \mu}^{\Omega_D} Tf, Tg \rangle, \quad f, g \in H^{1/2}(\partial D),$$

and denote its restriction to $\mathcal{X}_{\tilde{\Gamma}}(\partial D) \times \mathcal{X}_{\tilde{\Gamma}}(\partial D)$ by $\mathcal{T}_{\lambda, \mu}^{\tilde{\Gamma}}$, where the trace spaces are defined in Section 2.1.

Theorem 1.1. *Let D and Ω_D be as above, and put $\mu_j(r, \theta, z) = m_j(r, z)$ for $j = 1, 2$.*

- (i) *Suppose that D is simply connected, $m_j \in L^\infty(D)$, $\lambda_j \in L^\infty(\Omega_D)$, and both pairs satisfy (1.1). If their full torsional forms satisfy*

$$\mathcal{T}_{\lambda_1, \mu_1}(f, g) = \mathcal{T}_{\lambda_2, \mu_2}(f, g) \quad \text{for every } f, g \in H^{1/2}(\partial D),$$

then $\mu_1 = \mu_2$ almost everywhere in Ω_D . For fixed m , the entire torsional form is independent of the admissible first parameter λ , so this conclusion does not determine λ .

- (ii) *Suppose instead that m_j and λ_j are smooth in neighborhoods of $\overline{D}$ and $\overline{\Omega_D}$, respectively, and that both pairs satisfy (1.1). Simple connectedness of D is not required in this part. For any nonempty relatively open arc $\tilde{\Gamma}$,*

$$\mathcal{T}_{\lambda_1, \mu_1}^{\tilde{\Gamma}} = \mathcal{T}_{\lambda_2, \mu_2}^{\tilde{\Gamma}} \implies \mu_1 = \mu_2 \text{ on } \overline{\Omega_D}.$$

If the full elastic maps are equal, then $(\lambda_1, \mu_1) = (\lambda_2, \mu_2)$ on $\overline{\Omega_D}$.

In both parts, the first Lamé parameters may be non-axisymmetric.

Theorem 1.1 is based on an exact weak reduction of the torsional problem to a planar conductivity equation with coefficient $r^3 m$. Since D stays away from the rotation axis, this coefficient is uniformly elliptic. The reduction identifies full and partial torsional data with the corresponding conductivity data, allowing us to apply the planar uniqueness results.

1.2.2. *Separation of variables.* Let $G \subset \mathbb{R}^2$ be a bounded connected C^∞ domain, let $L > 0$, and set

$$\Omega = G \times (0, L), \quad x = (y, t).$$

Here t is a spatial coordinate. The product domain is Lipschitz, and its boundary is smooth away from the edges $\partial G \times \{0, L\}$. Fix a nonempty relatively open arc $\Sigma \subset \partial G$ and set

$$\Gamma_0 = G \times \{0\}, \quad \Gamma_\Sigma = \Sigma \times (0, L), \quad \Gamma = \Gamma_0 \cup \Gamma_\Sigma.$$

Theorem 1.2. *Let Ω , Σ , and Γ be as above. Let (λ_j, μ_j) , $j = 1, 2$, be smooth in a common neighborhood of $\overline{\Omega}$ and satisfy (1.1). Suppose that one of the following representations holds for both candidates on $\overline{G} \times [0, L]$:*

- (i) *Multiplicative separation:*

$$\mu_j(y, t) = \alpha_j(y) \beta_j(t), \quad j = 1, 2,$$

where $\alpha_j \in C^\infty(\overline{G})$ and $\beta_j \in C^\infty([0, L])$ are positive and both are unknown.

- (ii) *Additive separation:*

$$\mu_j(y, t) = a_j(y) + b_j(t), \quad j = 1, 2,$$

where $a_j \in C^\infty(\overline{G})$ and $b_j \in C^\infty([0, L])$ are both unknown. Neither summand is required to be positive separately; positivity of their sum is included in (1.1).

In either case,

$$\Lambda_{\lambda_1, \mu_1}^\Gamma = \Lambda_{\lambda_2, \mu_2}^\Gamma \implies \mu_1 = \mu_2 \text{ on } \overline{\Omega}.$$

If the full elastic Dirichlet-to-Neumann maps are equal, then

$$(\lambda_1, \mu_1) = (\lambda_2, \mu_2) \text{ on } \overline{\Omega}.$$

No structural condition is imposed on either λ_j .

Remark 1.1. Under either data equality in Theorem 1.2, the factors in case (i) are determined up to $(\alpha, \beta) \mapsto (c\alpha, c^{-1}\beta)$, where $c > 0$. In case (ii), the summands are determined up to $(a, b) \mapsto (a + c, b - c)$, where $c \in \mathbb{R}$. The normalizations $\beta(0) = 1$ and $b(0) = 0$ give individual uniqueness in the respective cases.

Remark 1.2. The multiplicative representation includes translation-invariant shear moduli when $\beta \equiv 1$ and layered shear moduli when α is constant. More generally, both factors or both summands may vary smoothly with arbitrary oscillation.

The additive class is not a special case of the multiplicative class. For example, $F(y, t) = 2 + y_1 + t$ is positive on the unit disk times $(0, 1)$ and is additively separated. Every smooth product satisfies $F\partial_{y_1 t}F - (\partial_{y_1}F)(\partial_t F) = 0$, whereas this expression equals -1 for the displayed additive example.

1.2.3. Directional quasianalyticity. Quasianalyticity preserves the identity principle beyond real-analytic functions, providing a natural route to uniqueness in inverse boundary problems [2, 4]. We use the following one-dimensional Denjoy–Carleman class.

Definition 1.1. Let $M = (M_k)_{k \geq 0}$ be a nondecreasing positive sequence with $M_0 = 1$ and

$$M_k^2 \leq M_{k-1}M_{k+1} \quad (k \geq 1), \quad \sum_{k=0}^{\infty} \frac{M_k}{(k+1)M_{k+1}} = \infty. \quad (1.5)$$

For a connected open interval $J \subset \mathbb{R}$, the *Denjoy–Carleman class of Roumieu type* $C^{\{M\}}(J)$ consists of the functions $q \in C^\infty(J)$ such that, for every compact $K \Subset J$, there are $C, R > 0$ with

$$\sup_{s \in K} |q^{(k)}(s)| \leq CR^k k! M_k, \quad k \geq 0.$$

The constants may depend on q and K , but not on k .

By the Denjoy–Carleman theorem [12, 14], this linear class is quasianalytic: for every $q \in C^{\{M\}}(J)$ and $s_* \in J$,

$$q^{(k)}(s_*) = 0 \text{ for all integers } k \geq 0 \implies q = 0 \text{ on } J. \quad (1.6)$$

Theorem 1.3. Let $\Omega \subset \mathbb{R}^3$ be bounded, connected, and C^∞ . Let λ_j, μ_j , $j = 1, 2$, be smooth in a common open neighborhood U of $\bar{\Omega}$ and satisfy (1.1). Fix $e \in S^2$ and, for $y \in e^\perp$, define

$$I_y = \{s \in \mathbb{R} : y + se \in \Omega\}.$$

For each nonempty connected component (a, b) of I_y , suppose there is a connected open interval $J \supset [a, b]$, with $y + Je \subset U$, such that the two functions

$$s \mapsto \mu_j(y + se), \quad s \in J, \quad j = 1, 2,$$

belong to a common class $C^{\{M\}}(J)$ as in Definition 1.1, with M prescribed independently of the candidates. The weight may vary between components; the defining constants need not be uniform across lines. Then equality of the full elastic Dirichlet-to-Neumann maps implies

$$(\lambda_1, \mu_1) = (\lambda_2, \mu_2) \quad \text{on } \bar{\Omega}.$$

No quasianalyticity or other structural assumption is imposed on λ_j , and no convexity of Ω is required.

Theorem 1.3 uses quasianalyticity along lines parallel to the fixed direction e . On each connected component of a line section, boundary determination provides all derivatives of μ at an endpoint, and quasianalyticity determines μ throughout that component. Since this argument applies to each component separately, the domain need not be convex.

Remark 1.3. Theorem 1.3 extends to shear moduli that are quasianalytic along a prescribed family of smooth curves. When these curves start on the observed boundary and cover Ω , localized boundary data determine μ throughout Ω , while full boundary data determine both Lamé parameters; see Proposition 5.1.

Remark 1.4. Theorem 1.3 also holds with $C^{\{M\}}(J)$ replaced by any prescribed linear subspace $\mathcal{Q}(J) \subset C^\infty(J)$ satisfying (1.6) for every $q \in \mathcal{Q}(J)$ and $s_* \in J$.

Remark 1.5. Theorem 1.3 extends uniqueness for real-analytic Lamé parameters to a strictly larger class of smooth coefficients. The shear modulus need only be quasianalytic in one fixed direction, allowing classes strictly larger than the real-analytic class [14]. No analyticity is required in the transverse variables or for λ .

1.3. Organization of the paper. The rest of the paper is organized as follows. Section 2 gives the weak formulation, boundary determination, and uniqueness for a known shear modulus. Section 3 proves Theorem 1.1 through the torsional reduction to planar conductivity. Sections 4 and 5 prove Theorems 1.2 and 1.3, respectively.

2. PRELIMINARIES

We collect the weak formulation and the results shared by the proofs of the main theorems.

2.1. Weak formulation and boundary measurements. Throughout the paper, smoothness on a closed set means restriction of a C^∞ function defined on an open neighborhood of that set. For smooth coefficients, (1.1) holds on the closed domain by continuity.

For bounded coefficients satisfying (1.1), define the elastic energy form by

$$\mathcal{B}_{\lambda,\mu}^\Omega(u, v) = \int_\Omega \left[\lambda(\operatorname{div} u)(\operatorname{div} v) + 2\mu \varepsilon(u) : \varepsilon(v) \right] dx, \quad u, v \in H^1(\Omega)^3. \quad (2.1)$$

The colon denotes the Frobenius product $A : B = \sum_{i,k=1}^3 A_{ik} B_{ik}$. When the domain is clear, its superscript is omitted. The trace theorem, Korn's inequality, and the Lax–Milgram theorem give the unique weak solution u_f for every $f \in H^{1/2}(\partial\Omega)^3$ [3]. The full Dirichlet-to-Neumann map is defined by

$$\langle \Lambda_{\lambda,\mu}^\Omega f, g \rangle = \mathcal{B}_{\lambda,\mu}^\Omega(u_f, V_g), \quad V_g|_{\partial\Omega} = g. \quad (2.2)$$

The right-hand side is independent of the extension V_g , since $\mathcal{B}_{\lambda,\mu}^\Omega(u_f, \phi) = 0$ for every $\phi \in H_0^1(\Omega)^3$. This weak definition applies to bounded measurable coefficients on Lipschitz domains, including product domains with edges. In the smooth setting, it agrees with the traction $\sigma_{\lambda,\mu}(u_f)\nu$.

For a relatively open set Γ in the smooth part of $\partial\Omega$, set

$$\mathcal{X}_\Gamma(\partial\Omega) := \overline{C_c^\infty(\Gamma)}^{H^{1/2}(\partial\Omega)}.$$

The localized boundary map is the bilinear form

$$\langle \Lambda_{\lambda,\mu}^\Gamma f, g \rangle := \langle \Lambda_{\lambda,\mu}^\Omega f, g \rangle, \quad f, g \in \mathcal{X}_\Gamma(\partial\Omega)^3.$$

Thus both the displacement f and the traction test g vanish almost everywhere on $\partial\Omega \setminus \Gamma$. Only these pairings are measured; the traction need not vanish on $\partial\Omega \setminus \Gamma$.

All forms are extended bilinearly to complex arguments. Equality for real inputs therefore implies equality for complex inputs; the sesquilinear convention used in some references is recovered by conjugating the second argument.

2.2. Local boundary determination. The following lemma expresses local boundary determination in Cartesian derivatives. For full data, continuity extends the identities to the edges of product domains.

Lemma 2.1. *Let Ω be a bounded connected domain with C^∞ boundary, or a product domain $G \times (0, L)$ with the geometry specified before Theorem 1.2. Let (λ_j, μ_j) , $j = 1, 2$, be smooth in a common neighborhood of $\bar{\Omega}$ and satisfy (1.1). Let W be a nonempty relatively open subset of the smooth part of $\partial\Omega$. If $\Lambda_{\lambda_1, \mu_1}^W = \Lambda_{\lambda_2, \mu_2}^W$, then*

$$D^\eta(\lambda_1 - \lambda_2)(p) = D^\eta(\mu_1 - \mu_2)(p) = 0 \quad (p \in W, \eta \in \mathbb{N}_0^3),$$

where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and D^η denotes a Cartesian derivative. If the full elastic maps are equal, these identities hold on all of $\partial\Omega$, including the edges of a product domain.

Proof. Step 1: localization and boundary values. Work in a smooth boundary patch contained in W . Choose a smaller patch $W' \Subset W$. Let χ be a smooth cutoff supported in W , with $\chi = 1$ on W' . For arbitrary $f, g \in H^{1/2}(\partial\Omega)^3$,

$$\langle (\Lambda_{\lambda_1, \mu_1}^\Omega - \Lambda_{\lambda_2, \mu_2}^\Omega)(\chi f), \chi g \rangle = 0.$$

Here χf and χg belong to $\mathcal{X}_W(\partial\Omega)^3$ by approximation inside the smooth chart. Thus, as operators from $H^{1/2}(\partial\Omega)^3$ to $H^{-1/2}(\partial\Omega)^3$,

$$\chi(\Lambda_{\lambda_1, \mu_1}^\Omega - \Lambda_{\lambda_2, \mu_2}^\Omega)\chi = 0.$$

On W' , the localized maps are classical pseudodifferential operators of order one [10, 13]. For a product domain, their boundary parametrices are constructed inside a smooth face. Local boundary regularity makes the correction from the rest of the boundary smoothing on W' . Thus the edges do not affect the local symbols. In any common boundary chart and frame, the homogeneous symbol terms satisfy

$$p_{1-k}^{(1)}(\xi', \zeta') = p_{1-k}^{(2)}(\xi', \zeta'), \quad k \geq 0, \quad \zeta' \neq 0,$$

over W' , since $\chi = 1$ there.

The local reconstruction in [9, Theorem 1.1 and Section 4; 10] first determines the boundary values of both parameters.

Step 2: higher normal derivatives. Higher normal derivatives are recovered recursively from the lower orders [13].

Fix a smooth boundary normal chart $x = F(\xi', s)$, with s increasing into Ω . Keep the displacement components in the original Cartesian basis. Put $J = DF$ and $S = J^{-1}$. For $q = \lambda, \mu$, set $\hat{q}_j = q_j \circ F$ and $\delta\hat{q} = \hat{q}_1 - \hat{q}_2$. If $C_{ik\ell m}^{(j)}$ is the Cartesian elastic tensor, changing variables in the energy gives the coefficient

$$\hat{C}_{ialb}^{(j)} = |\det J| \sum_{k,m=1}^3 C_{ik\ell m}^{(j)}(F(\xi', s)) S_{ak} S_{bm} = A_{ialb} \hat{\lambda}_j + B_{ialb} \hat{\mu}_j,$$

where the tensor indices i, a, ℓ, b, k, m range from 1 to 3, and the known geometric tensors are

$$A_{ialb} = |\det J| S_{ai} S_{bl},$$

$$B_{ialb} = |\det J| \left(\delta_{il} \sum_{k=1}^3 S_{ak} S_{bk} + S_{al} S_{bi} \right).$$

Indeed, $C_{ik\ell m}^{(j)} = \lambda_j \delta_{ik} \delta_{\ell m} + \mu_j (\delta_{i\ell} \delta_{km} + \delta_{im} \delta_{k\ell})$. The factor $|\det J|$ accounts for the volume change. The transformed tensor need not be isotropic.

Let $V' = \{\xi' : F(\xi', 0) \in W'\}$. For $k \geq 1$, suppose inductively that

$$\partial_s^\ell \delta \hat{\lambda}(\xi', 0) = \partial_s^\ell \delta \hat{\mu}(\xi', 0) = 0, \quad \xi' \in V', \quad 0 \leq \ell < k.$$

The product rule, applied to the preceding tensor identity, gives

$$\partial_s^k (\hat{C}^{(1)} - \hat{C}^{(2)}) = \sum_{\ell=0}^k \binom{k}{\ell} [(\partial_s^{k-\ell} A) \partial_s^\ell \delta \hat{\lambda} + (\partial_s^{k-\ell} B) \partial_s^\ell \delta \hat{\mu}].$$

At $s = 0$, every term with $\ell < k$ is zero by induction. Therefore the only undetermined terms at this order are

$$\partial_s^k (\hat{C}^{(1)} - \hat{C}^{(2)})|_{s=0} = A(\xi', 0) \partial_s^k \delta \hat{\lambda}(\xi', 0) + B(\xi', 0) \partial_s^k \delta \hat{\mu}(\xi', 0).$$

To choose a common reference operator, define

$$P_{k-1}^q(\xi', s) = \sum_{\ell=0}^{k-1} \frac{s^\ell}{\ell!} \partial_s^\ell \hat{q}_1(\xi', 0), \quad q = \lambda, \mu.$$

The induction hypothesis gives the same polynomials for the second candidate. On a sufficiently thin neighborhood V of $\overline{W'}$, continuity gives

$$P_{k-1}^\mu \circ F^{-1} \geq c_*/2, \quad (3P_{k-1}^\lambda + 2P_{k-1}^\mu) \circ F^{-1} \geq c_*/2.$$

Choose a constant strongly convex pair (λ_*, μ_*) . Take $\rho \in C_c^\infty(V)$ with $0 \leq \rho \leq 1$ and $\rho = 1$ near $\overline{W'}$. Using this same cutoff for both parameters, set

$$q_{\text{ref}} = \rho(P_{k-1}^q \circ F^{-1}) + (1 - \rho)q_*, \quad q = \lambda, \mu.$$

The products with ρ are extended by zero outside V . With $c_0 = \min\{c_*/2, \mu_*, 3\lambda_* + 2\mu_*\} > 0$, the affine inequalities give

$$\mu_{\text{ref}} \geq c_0, \quad 3\lambda_{\text{ref}} + 2\mu_{\text{ref}} \geq c_0.$$

Moreover,

$$\partial_s^\ell (q_{\text{ref}} \circ F)(\xi', 0) = \partial_s^\ell \hat{q}_j(\xi', 0), \quad \xi' \in V', \quad 0 \leq \ell < k, \quad j = 1, 2.$$

To apply the symbol recursion, express displacement and traction in the boundary coordinate frame used in [13]. The change from Cartesian components and the boundary density are determined by F and are common to both candidates. The localized operators therefore remain equal after this change. Fix $\xi' \in V'$ and write

$$\lambda_0 = \hat{\lambda}_1(\xi', 0) = \hat{\lambda}_2(\xi', 0), \quad \mu_0 = \hat{\mu}_1(\xi', 0) = \hat{\mu}_2(\xi', 0).$$

In the recursion at symbol order $1 - k$, the entries with one tangential and one normal index and the entry with two normal indices determine two scalar combinations of the k th normal derivatives. Compare each candidate's recursion with that for $(\lambda_{\text{ref}}, \mu_{\text{ref}})$ in the same coordinate frame. The lower-order terms depend only on the geometry and the common

derivatives of normal order less than k , so they cancel in each comparison. Subtracting the resulting identities for the two candidates removes the reference contribution, and rescaling gives

$$\begin{pmatrix} \mu_0 & -(2\lambda_0 + 3\mu_0) \\ \mu_0^2 & \lambda_0^2 + 4\lambda_0\mu_0 + 6\mu_0^2 \end{pmatrix} \begin{pmatrix} \partial_s^k \delta \hat{\lambda}(\xi', 0) \\ \partial_s^k \delta \hat{\mu}(\xi', 0) \end{pmatrix} = 0.$$

The rescaling is valid because $\lambda_0 + 2\mu_0 > 0$ and $\lambda_0 + 3\mu_0 > 0$. The determinant of this matrix is

$$\mu_0(\lambda_0^2 + 4\lambda_0\mu_0 + 6\mu_0^2) + (2\lambda_0 + 3\mu_0)\mu_0^2 = \mu_0(\lambda_0 + 3\mu_0)^2 > 0.$$

Hence

$$\partial_s^k \delta \hat{\lambda}(\xi', 0) = \partial_s^k \delta \hat{\mu}(\xi', 0) = 0, \quad \xi' \in V'.$$

Induction proves equality of all normal derivatives. The neighborhood V may depend on k .

Step 3: Cartesian derivatives and edges. These equalities hold as identities of smooth functions of ξ' . Tangential differentiation gives

$$\begin{aligned} \partial_{\xi'}^\eta \partial_s^k \hat{\lambda}_1(\xi', 0) &= \partial_{\xi'}^\eta \partial_s^k \hat{\lambda}_2(\xi', 0), \\ \partial_{\xi'}^\eta \partial_s^k \hat{\mu}_1(\xi', 0) &= \partial_{\xi'}^\eta \partial_s^k \hat{\mu}_2(\xi', 0), \end{aligned} \tag{2.3}$$

for every $\xi' \in V'$, $\eta \in \mathbb{N}_0^2$, and $k \in \mathbb{N}_0$. These are derivatives of the compositions with the fixed chart F . To return to Cartesian derivatives, write $\xi_3 = s$ and apply

$$(\partial_{x_i} q) \circ F = \sum_{a=1}^3 S_{ai} \partial_{\xi_a} (q \circ F).$$

Iteration gives, for $\delta q = q_1 - q_2$,

$$(D_x^\alpha \delta q) \circ F = \sum_{|\beta| \leq |\alpha|} c_{\alpha\beta} D_\xi^\beta \delta \hat{q}.$$

The coefficients $c_{\alpha\beta}$ are smooth and depend only on the chart. Every term on the right vanishes at $s = 0$ by (2.3). Thus all Cartesian boundary derivatives agree on W' .

Each point of W lies in a smaller patch of this kind. On a product domain, charts around interior points of the boundary faces avoid the edges. The same argument therefore applies on each smooth face. For full data, let p be an edge point. Take points p_n on a smooth face with $p_n \rightarrow p$. Continuity of the ambient derivatives gives

$$D^\alpha (q_1 - q_2)(p) = \lim_{n \rightarrow \infty} D^\alpha (q_1 - q_2)(p_n) = 0, \quad q = \lambda, \mu.$$

The proof is complete. □

2.3. Uniqueness for a known shear modulus. We first extend the coefficients to a ball while preserving the common shear modulus and equality of the boundary maps.

Lemma 2.2. *Let Ω be a bounded connected domain with C^∞ boundary, or a product domain $G \times (0, L)$ with the geometry specified before Theorem 1.2. Let (λ_j, μ_j) , $j = 1, 2$, be smooth in a common neighborhood of $\overline{\Omega}$ and satisfy (1.1). Suppose that*

$$\Lambda_{\lambda_1, \mu_1}^\Omega = \Lambda_{\lambda_2, \mu_2}^\Omega, \quad \mu_1 = \mu_2 \quad \text{in } \Omega.$$

Then there exist a ball B with $\bar{\Omega} \Subset B$ and smooth strongly convex pairs $(\tilde{\lambda}_j, \tilde{\mu}_j)$ on $\bar{B}$ such that

$$\begin{aligned} (\tilde{\lambda}_j, \tilde{\mu}_j)|_{\Omega} &= (\lambda_j, \mu_j), \quad j = 1, 2, \\ (\tilde{\lambda}_1, \tilde{\mu}_1) &= (\tilde{\lambda}_2, \tilde{\mu}_2) \quad \text{in } B \setminus \Omega, \\ \tilde{\mu}_1 &= \tilde{\mu}_2 \quad \text{in } B, \\ \Lambda_{\tilde{\lambda}_1, \tilde{\mu}_1}^B &= \Lambda_{\tilde{\lambda}_2, \tilde{\mu}_2}^B. \end{aligned}$$

Both pairs equal the same constant strongly convex pair near ∂B .

Proof. Step 1: common extensions. Lemma 2.1, including its conclusion at the edges, gives

$$D^\eta(\lambda_1 - \lambda_2) = D^\eta(\mu_1 - \mu_2) = 0 \quad \text{on } \partial\Omega, \quad \eta \in \mathbb{N}_0^3, \quad (2.4)$$

where D^η denotes a Cartesian derivative. Let $q = \lambda_2 - \lambda_1$. For each multi-index η , let Q_η be the zero extension of $D^\eta q$ outside Ω . Put $Q = Q_0$. Fix $p \in \partial\Omega$. Taylor's formula in the ambient neighborhood and (2.4) give

$$|Q_\eta(p+h) - Q_\eta(p)| \leq C_{p,\eta}|h|^2 \quad \text{for sufficiently small } |h|.$$

Thus Q_η is continuous and differentiable at p , with derivative zero. Away from $\partial\Omega$, differentiation follows from the definition. Hence

$$\partial_i Q_\eta = Q_{\eta+e_i}, \quad i = 1, 2, 3,$$

where e_i is the i th coordinate multi-index. Each function on the right is continuous. Induction yields

$$Q \in C_c^\infty(\mathbb{R}^3), \quad D^\eta Q = Q_\eta.$$

The argument uses ambient derivatives and also applies at the edges.

Choose a bounded neighborhood U_0 of $\bar{\Omega}$ on which

$$\mu_1 \geq c_*/2, \quad 3\lambda_1 + 2\mu_1 \geq c_*/2.$$

Take $\chi \in C_c^\infty(U_0)$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near $\bar{\Omega}$. Choose a constant strongly convex pair (λ_*, μ_*) . Set

$$\begin{aligned} (\tilde{\lambda}_1, \tilde{\mu}_1) &= \chi(\lambda_1, \mu_1) + (1-\chi)(\lambda_*, \mu_*), \\ (\tilde{\lambda}_2, \tilde{\mu}_2) &= (\tilde{\lambda}_1 + Q, \tilde{\mu}_1), \end{aligned}$$

where the products with χ are extended by zero outside U_0 . The construction gives

$$\begin{aligned} (\tilde{\lambda}_j, \tilde{\mu}_j)|_{\Omega} &= (\lambda_j, \mu_j), \quad j = 1, 2, \\ (\tilde{\lambda}_2, \tilde{\mu}_2) &= (\tilde{\lambda}_1, \tilde{\mu}_1) \quad \text{outside } \Omega. \end{aligned}$$

The first pair is a convex combination of strongly convex pairs. The matching relations give the same bounds for the second pair. With $c_0 = \min\{c_*/2, \mu_*, 3\lambda_* + 2\mu_*\} > 0$, we obtain

$$\tilde{\mu}_j \geq c_0, \quad 3\tilde{\lambda}_j + 2\tilde{\mu}_j \geq c_0, \quad j = 1, 2.$$

Choose a ball B such that $\bar{\Omega} \cup \text{supp } \chi \Subset B$. Both pairs equal (λ_*, μ_*) near ∂B . Moreover,

$$\tilde{\mu}_1 = \tilde{\mu}_2 \quad \text{in } B.$$

These extensions need not preserve axisymmetry, separation, or quasianalyticity.

Step 2: transfer of boundary data. To compare the boundary maps, fix $F \in H^{1/2}(\partial B)^3$. Let $U_1 \in H^1(B)^3$ solve the first extended equation with trace F . Put $f = \text{Tr}_{\partial\Omega}(U_1|_{\Omega})$. Extending tests in $H_0^1(\Omega)^3$ by zero shows that $U_1|_{\Omega}$ solves the first original equation. Let v_2 solve the

second original equation with trace f . Set $w = v_2 - U_1|_\Omega \in H_0^1(\Omega)^3$. Let E_0 denote extension by zero, applied componentwise. Approximation by compactly supported smooth functions gives

$$E_0 w \in H_0^1(B)^3, \quad \nabla(E_0 w) = E_0(\nabla w).$$

Define

$$U_2 := U_1 + E_0 w = \begin{cases} v_2, & \text{in } \Omega, \\ U_1, & \text{in } B \setminus \bar{\Omega} \end{cases} \in H^1(B)^3.$$

Its weak gradient is the corresponding piecewise gradient. Its outer trace is F . For $\Phi \in H_0^1(B)^3$, put $\varphi = \text{Tr}_{\partial\Omega}(\Phi|_\Omega)$. Splitting the energies over Ω and its exterior and using equality of the original maps gives

$$\begin{aligned} \mathcal{B}_{\lambda_2, \mu_2}^B(U_2, \Phi) &= \mathcal{B}_{\lambda_1, \mu_1}^B(U_1, \Phi) + \left\langle (\Lambda_{\lambda_2, \mu_2}^\Omega - \Lambda_{\lambda_1, \mu_1}^\Omega) f, \varphi \right\rangle \\ &= 0. \end{aligned}$$

Hence U_2 is the second extended solution. Given $H \in H^{1/2}(\partial B)^3$, choose $V_H \in H^1(B)^3$ with trace H . Take a smooth cutoff ζ equal to zero near $\bar{\Omega}$ and to one near ∂B . Then $W = \zeta V_H$ has trace H and vanishes near $\bar{\Omega}$. The solutions and coefficients agree on $\text{supp } W$. Therefore

$$\begin{aligned} \left\langle \Lambda_{\lambda_2, \mu_2}^B F, H \right\rangle &= \mathcal{B}_{\lambda_2, \mu_2}^B(U_2, W) \\ &= \mathcal{B}_{\lambda_1, \mu_1}^B(U_1, W) = \left\langle \Lambda_{\lambda_1, \mu_1}^B F, H \right\rangle. \end{aligned}$$

Since F and H are arbitrary,

$$\Lambda_{\lambda_1, \mu_1}^B = \Lambda_{\lambda_2, \mu_2}^B.$$

No connectedness assumption on $B \setminus \bar{\Omega}$ is needed.

The proof is complete. $\square$

The weighted estimate gives uniqueness of the first Lamé parameter when the shear modulus is known [5, 6].

Proposition 2.3. *Let Ω be a bounded connected domain with C^∞ boundary, or a product domain $G \times (0, L)$ with the geometry specified before Theorem 1.2. Let (λ_j, μ_j) , $j = 1, 2$, be smooth in a common neighborhood of $\bar{\Omega}$ and satisfy (1.1). If*

$$\Lambda_{\lambda_1, \mu_1}^\Omega = \Lambda_{\lambda_2, \mu_2}^\Omega \quad \text{and} \quad \mu_1 = \mu_2 \quad \text{in } \Omega,$$

then $\lambda_1 = \lambda_2$ on $\bar{\Omega}$.

Proof. Lemma 2.2 gives smooth strongly convex extensions to a ball B with a common shear modulus and equal full boundary maps. Both extended pairs equal the same constant pair near ∂B . For either extended pair, strong convexity gives

$$\tilde{\lambda} + \tilde{\mu} = \frac{3\tilde{\lambda} + 2\tilde{\mu} + \tilde{\mu}}{3} > 0, \quad \tilde{\lambda} + 2\tilde{\mu} = \frac{3\tilde{\lambda} + 2\tilde{\mu} + 4\tilde{\mu}}{3} > 0.$$

Together with $\tilde{\mu} > 0$, these are the positivity conditions required by the elasticity reduction. For these extensions, [5, Theorem 2] gives constants $C, \tau_0 > 0$ such that

$$\|\tilde{\lambda}_1 + \tilde{\mu}_1 - \tilde{\lambda}_2 - \tilde{\mu}_2\|_\tau \leq \frac{C}{\tau} \|\tilde{\mu}_1 - \tilde{\mu}_2\|_\tau, \quad \tau > \tau_0, \quad (2.5)$$

where C is independent of τ . Both C and τ_0 may depend on the extended coefficients and B . The weighted norm is

$$\|q\|_\tau^2 = \int_B |q(x)|^2 e^{2\tau|x|^2} dx.$$

Estimate (2.5) requires no smallness of $\nabla \tilde{\mu}$. The gradient restriction in [5, Theorem 1] belongs to a separate perturbative uniqueness result.

Fix $\tau > \tau_0$. Using the common shear modulus in (2.5), we obtain

$$\|\tilde{\lambda}_1 - \tilde{\lambda}_2\|_\tau \leq \frac{C}{\tau} \|\tilde{\mu}_1 - \tilde{\mu}_2\|_\tau = 0.$$

Positivity of the weight gives equality almost everywhere in B . Smoothness then yields $\lambda_1 = \lambda_2$ on $\overline{\Omega}$.

The proof is complete. $\square$

3. AXISYMMETRIC SHEAR MODULI

Torsional measurements reduce the recovery of an axisymmetric shear modulus to a planar conductivity problem. We use this reduction to prove Theorem 1.1.

3.1. Planar conductivity uniqueness. The following proposition gives the planar uniqueness results needed to determine the shear modulus in Theorem 1.1. Proposition 3.2 connects the elastic measurements with conductivity data.

Let $D \subset \mathbb{R}^2$ be a bounded connected domain with C^∞ boundary. For a real conductivity $\gamma \in L^\infty(D)$ satisfying $0 < c_\gamma \leq \gamma \leq C_\gamma < \infty$ almost everywhere, let $v_f \in H^1(D)$ solve

$$\operatorname{div}(\gamma \nabla v_f) = 0 \quad \text{in } D, \quad v_f|_{\partial D} = f \in H^{1/2}(\partial D).$$

The scalar Dirichlet-to-Neumann map $\Lambda_\gamma^c : H^{1/2}(\partial D) \rightarrow H^{-1/2}(\partial D)$ is defined by

$$\langle \Lambda_\gamma^c f, g \rangle = \int_D \gamma \nabla v_f \cdot \nabla w_g dx, \quad f, g \in H^{1/2}(\partial D),$$

where $w_g \in H^1(D)$ is any extension of g .

Proposition 3.1. *Let D be as above and let $\gamma_j \in L^\infty(D)$, $j = 1, 2$, be real valued, with*

$$0 < c \leq \gamma_j \leq C < \infty \quad \text{almost everywhere in } D.$$

(i) *If D is simply connected, then*

$$\Lambda_{\gamma_1}^c = \Lambda_{\gamma_2}^c \implies \gamma_1 = \gamma_2 \quad \text{almost everywhere in } D.$$

(ii) *Suppose that $\gamma_j \in C^\infty(\overline{D})$ and let $\tilde{\Gamma} \subset \partial D$ be any nonempty relatively open arc. If*

$$\langle (\Lambda_{\gamma_1}^c - \Lambda_{\gamma_2}^c) f, g \rangle = 0 \quad \text{for all } f, g \in \mathcal{X}_{\tilde{\Gamma}}(\partial D),$$

then $\gamma_1 = \gamma_2$ on $\overline{D}$. No simple connectedness assumption is required in this part.

Proof. Part (i) is the planar conductivity uniqueness theorem in [1, Theorem 1]. For part (ii), take $f \in H^{1/2}(\partial D)$ with $\operatorname{supp} f \Subset \tilde{\Gamma}$. Boundary mollification gives $f_n \in C_c^\infty(\tilde{\Gamma})$ with $f_n \rightarrow f$ in $H^{1/2}(\partial D)$. The assumed pairing identity and boundedness of the scalar maps imply

$$(\Lambda_{\gamma_1}^c f)|_{\tilde{\Gamma}} = (\Lambda_{\gamma_2}^c f)|_{\tilde{\Gamma}} \quad \text{in } \mathcal{D}'(\tilde{\Gamma}).$$

The uniqueness result for inputs and measurements on the same arc [7, Corollary 1.1] therefore gives $\gamma_1 = \gamma_2$ in D , and continuity extends the equality to $\overline{D}$. $\square$

3.2. The exact torsional reduction. For bounded measurable coefficients, the torsional lift must satisfy the three-dimensional weak equation against arbitrary vector tests. The angular average of these tests reduces the elastic form to the planar conductivity form. The next proposition gives this reduction and the resulting boundary pairing.

Proposition 3.2. *Let D and Ω_D be as in Theorem 1.1. Let $m \in L^\infty(D)$ be bounded below by a positive constant, set $\mu(r, \theta, z) = m(r, z)$, and let $\lambda \in L^\infty(\Omega_D)$ satisfy (1.1) together with μ . Define*

$$\gamma(r, z) = r^3 m(r, z).$$

The map $T : f \mapsto r f e_\theta$ is bounded from $H^{1/2}(\partial D)$ to $H^{1/2}(\partial\Omega_D)^3$, and maps $\mathcal{X}_\Gamma(\partial D)$ into $\mathcal{X}_\Gamma(\partial\Omega_D)^3$. If v_f is the scalar conductivity solution with coefficient γ and trace f , then $u_{Tf} = r v_f e_\theta$ is the weak elastic solution with trace Tf . Moreover,

$$\langle \Lambda_{\lambda, \mu}^{\Omega_D} T f, T g \rangle = 2\pi \langle \Lambda_\gamma^c f, g \rangle, \quad f, g \in H^{1/2}(\partial D). \quad (3.1)$$

Proof. Step 1: the lift and its boundary trace. The angle θ is understood modulo 2π . The cylindrical map identifies $D \times S^1$ with Ω_D . Write $e_r = (\cos \theta, \sin \theta, 0)$ and $e_z = (0, 0, 1)$. For a smooth scalar function v , set $Ev = r v e_\theta$. Direct differentiation gives

$$\partial_r(Ev) = (v + r \partial_r v) e_\theta, \quad \partial_z(Ev) = r \partial_z v e_\theta, \quad r^{-1} \partial_\theta(Ev) = -v e_r.$$

Consequently, for the standard H^1 norm,

$$\begin{aligned} \|Ev\|_{H^1(\Omega_D)^3}^2 &= 2\pi \int_D r \left(r^2 |v|^2 + |v + r \partial_r v|^2 \right. \\ &\quad \left. + r^2 |\partial_z v|^2 + |v|^2 \right) dr dz \leq C_D \|v\|_{H^1(D)}^2. \end{aligned}$$

The bounds $r_- \leq r \leq r_+$ justify the inequality. Density extends E to a bounded map from $H^1(D)$ to $H^1(\Omega_D)^3$. Since $E(C_c^\infty(D)) \subset C_c^\infty(\Omega_D)^3$, density also gives

$$E(H_0^1(D)) \subset H_0^1(\Omega_D)^3.$$

Let $R_D : H^{1/2}(\partial D) \rightarrow H^1(D)$ be a bounded trace right inverse. Define $Tf = \text{Tr}_{\partial\Omega_D}(ER_D f)$. The trace theorem and the bound for E give

$$\|Tf\|_{H^{1/2}(\partial\Omega_D)^3} \leq C_D \|R_D f\|_{H^1(D)} \leq C'_D \|f\|_{H^{1/2}(\partial D)}.$$

For any two trace right inverses $R_D^{(1)}$ and $R_D^{(2)}$,

$$(R_D^{(1)} - R_D^{(2)})f \in H_0^1(D), \quad \text{Tr}_{\partial\Omega_D} E(R_D^{(1)} - R_D^{(2)})f = 0.$$

Thus T is independent of the chosen right inverse. For smooth f , one has $Tf = r f e_\theta$. Density gives the stated formula for every $f \in H^{1/2}(\partial D)$. For $v \in H^1(D)$, the difference $v - R_D \text{Tr}_{\partial D} v$ belongs to $H_0^1(D)$. Hence

$$\text{Tr}_{\partial\Omega_D}(Ev) = T \text{Tr}_{\partial D} v, \quad v \in H^1(D).$$

The support of a smooth lifted trace lies in the corresponding rotation band:

$$T(C_c^\infty(\tilde{\Gamma})) \subset C_c^\infty(\Gamma)^3.$$

The bound for T and the definition of the localized trace spaces give

$$T(\mathcal{X}_\Gamma(\partial D)) \subset \mathcal{X}_\Gamma(\partial\Omega_D)^3.$$

Step 2: strain components and the averaged test. Let $v \in H^1(D)$ and put $u = rve_\theta$. For smooth v , the components in the orthonormal cylindrical frame (e_r, e_θ, e_z) are $u_r = u_z = 0$ and $u_\theta = rv$. Consequently

$$\operatorname{div} u = 0, \quad \varepsilon_{r\theta}(u) = \varepsilon_{\theta r}(u) = \frac{r}{2} \partial_r v, \quad \varepsilon_{z\theta}(u) = \varepsilon_{\theta z}(u) = \frac{r}{2} \partial_z v,$$

and all other strain components vanish. Approximation of v by smooth functions in $H^1(D)$ proves the same statements for weak derivatives. In particular, the term containing λ vanishes even if λ depends on θ .

Take an arbitrary vector test field $\Phi \in C_c^\infty(\Omega_D)^3$. Write its cylindrical components as $\Phi_r, \Phi_\theta, \Phi_z$. Define

$$P\Phi(r, z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{\Phi_\theta(r, \theta, z)}{r} d\theta.$$

For a smooth test field,

$$\begin{aligned} \partial_r(P\Phi) &= \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{\partial_r \Phi_\theta}{r} - \frac{\Phi_\theta}{r^2} \right) d\theta, \\ \partial_z(P\Phi) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\partial_z \Phi_\theta}{r} d\theta. \end{aligned}$$

Using Cauchy–Schwarz, $dx = r dr d\theta dz$, and the bounds on r , we obtain

$$\|P\Phi\|_{H^1(D)} \leq C_D \|\Phi\|_{H^1(\Omega_D)^3}.$$

The projection of $\operatorname{supp} \Phi$ is compact in D . Thus $P\Phi \in C_c^\infty(D)$. Hence P extends by density to a bounded map from $H_0^1(\Omega_D)^3$ to $H_0^1(D)$.

Step 3: the equation against arbitrary vector tests. The entries of the general strain tensor that pair with those of u are

$$\begin{aligned} \varepsilon_{r\theta}(\Phi) &= \frac{1}{2} \left(\partial_r \Phi_\theta - \frac{\Phi_\theta}{r} + \frac{1}{r} \partial_\theta \Phi_r \right), \\ \varepsilon_{z\theta}(\Phi) &= \frac{1}{2} \left(\partial_z \Phi_\theta + \frac{1}{r} \partial_\theta \Phi_z \right). \end{aligned}$$

Each off-diagonal strain entry occurs twice in the Frobenius product. Using the volume element $dx = r dr d\theta dz$ in (2.1) gives

$$\begin{aligned} \mathcal{B}_{\lambda, \mu}^{\Omega_D}(u, \Phi) &= \int_D \int_0^{2\pi} mr^2 \left[\partial_r v \left(\partial_r \Phi_\theta - \frac{\Phi_\theta}{r} + \frac{1}{r} \partial_\theta \Phi_r \right) \right. \\ &\quad \left. + \partial_z v \left(\partial_z \Phi_\theta + \frac{1}{r} \partial_\theta \Phi_z \right) \right] d\theta dr dz. \end{aligned}$$

Since $m \in L^\infty(D)$, $v \in H^1(D)$, and Φ is smooth with compact support, Fubini's theorem applies. Both m and v are independent of θ . The terms containing $\partial_\theta \Phi_r$ and $\partial_\theta \Phi_z$ therefore integrate to zero by periodicity. For the remaining terms,

$$\partial_r \Phi_\theta - \frac{\Phi_\theta}{r} = r \partial_r \left(\frac{\Phi_\theta}{r} \right), \quad \partial_z \Phi_\theta = r \partial_z \left(\frac{\Phi_\theta}{r} \right).$$

Hence

$$\mathcal{B}_{\lambda, \mu}^{\Omega_D}(rve_\theta, \Phi) = 2\pi \int_D \gamma \nabla v \cdot \nabla(P\Phi) dr dz.$$

Keep m fixed. The bounds for E and P , together with boundedness of the coefficients, give continuity on $H^1(D) \times H_0^1(\Omega_D)^3$. Density extends the identity to this space. For $v = v_f$, the scalar weak equation and Step 1 give

$$\begin{aligned}\mathcal{B}_{\lambda,\mu}^{\Omega_D}(Ev_f, \Phi) &= 2\pi \int_D \gamma \nabla v_f \cdot \nabla(P\Phi) dr dz = 0, \\ \text{Tr}_{\partial\Omega_D}(Ev_f) &= Tf.\end{aligned}$$

Here $\Phi \in H_0^1(\Omega_D)^3$ is arbitrary, and $P\Phi \in H_0^1(D)$ justifies the vanishing integral. Uniqueness of the Dirichlet problem gives $Ev_f = u_{Tf}$.

Step 4: the boundary pairing. Let $w \in H^1(D)$ have trace g . By Step 1,

$$W = Ew = rwe_\theta \in H^1(\Omega_D)^3, \quad \text{Tr}_{\partial\Omega_D} W = Tg.$$

The displayed strain formulas give

$$2m\varepsilon(rv_f e_\theta) : \varepsilon(rwe_\theta) = mr^2 \nabla v_f \cdot \nabla w.$$

Using (2.2) and integrating over Ω_D , we obtain

$$\begin{aligned}\langle \Lambda_{\lambda,\mu}^{\Omega_D} Tf, Tg \rangle &= \mathcal{B}_{\lambda,\mu}^{\Omega_D}(Ev_f, Ew) \\ &= 2\pi \int_D \gamma \nabla v_f \cdot \nabla w dr dz = 2\pi \langle \Lambda_\gamma^c f, g \rangle.\end{aligned}$$

This proves (3.1).

The proof is complete. $\square$

3.3. Completion of the proof.

Proof of Theorem 1.1. For each candidate define $\gamma_j = r^3 m_j$. In part (i), the radial bounds and the positive lower and upper bounds of m_j give constants $c, C > 0$ such that

$$cr_-^3 \leq \gamma_j \leq Cr_+^3 \quad \text{almost everywhere in } D.$$

Thus γ_j are uniformly positive bounded measurable planar conductivities. For these bounded strongly convex elastic pairs, Proposition 3.2 gives

$$2\pi \langle (\Lambda_{\gamma_1}^c - \Lambda_{\gamma_2}^c) f, g \rangle = \mathcal{T}_{\lambda_1, \mu_1}(f, g) - \mathcal{T}_{\lambda_2, \mu_2}(f, g) = 0$$

for every $f, g \in H^{1/2}(\partial D)$. Since both scalar traces are arbitrary, the full conductivity maps agree as operators from $H^{1/2}(\partial D)$ to $H^{-1/2}(\partial D)$. Since D is bounded, smooth, and simply connected, Proposition 3.1(i) gives $\gamma_1 = \gamma_2$ almost everywhere in D . Dividing by the known positive factor r^3 gives $m_1 = m_2$ almost everywhere in D . Since $dx = r dr d\theta dz$, it follows that $\mu_1 = \mu_2$ almost everywhere in Ω_D .

For fixed m , the scalar coefficient $\gamma = r^3 m$ and its boundary map are independent of λ . Thus (3.1) shows that the entire torsional form is independent of λ . In particular, the admissible choices $\lambda = 0$ and $\lambda = 1$ give the same torsional data. This proves both the recovery of μ and the stated nonuniqueness of λ in part (i).

In part (ii), smoothness of m_j and the radial bounds give $\gamma_j \in C^\infty(\overline{D})$ with uniform positive lower and upper bounds. Proposition 3.2 maps traces supported on $\tilde{\Gamma}$ to elastic traces supported on its rotated surface Γ . For every $f, g \in \mathcal{X}_{\tilde{\Gamma}}(\partial D)$, the same identity turns equality of the local torsional forms into

$$\langle (\Lambda_{\gamma_1}^c - \Lambda_{\gamma_2}^c) f, g \rangle = 0.$$

These pairings give the local conductivity data on the nonempty arc $\tilde{\Gamma}$. Proposition 3.1(ii) therefore yields $\gamma_1 = \gamma_2$ on $\overline{D}$. This application does not require D to be simply connected. Division by r^3 and continuity give $\mu_1 = \mu_2$ on $\overline{\Omega_D}$.

To prove the simultaneous recovery assertion in part (ii), suppose now that the full elastic maps agree. Restriction to lifted traces gives

$$\begin{aligned}\mathcal{T}_{\lambda_1, \mu_1}^{\tilde{\Gamma}}(f, g) &= \langle \Lambda_{\lambda_1, \mu_1}^{\Omega_D} T f, T g \rangle \\ &= \langle \Lambda_{\lambda_2, \mu_2}^{\Omega_D} T f, T g \rangle = \mathcal{T}_{\lambda_2, \mu_2}^{\tilde{\Gamma}}(f, g)\end{aligned}$$

for all $f, g \in \mathcal{X}_{\tilde{\Gamma}}(\partial D)$. The local conclusion just proved gives $\mu_1 = \mu_2$ on $\overline{\Omega_D}$. Since $\overline{D} \subset \{r > 0\}$, the cylindrical parametrization gives a bounded connected smooth domain Ω_D . Moreover, the lifts $\mu_j(x) = m_j(\sqrt{x_1^2 + x_2^2}, x_3)$ are smooth near $\overline{\Omega_D}$. Intersecting these neighborhoods with those of λ_j gives a common neighborhood on which both pairs are smooth. The two smooth strongly convex pairs therefore have a common shear modulus and equal full maps on Ω_D . Proposition 2.3 gives $\lambda_1 = \lambda_2$ on $\overline{\Omega_D}$, proving part (ii). No axisymmetry of λ_j is used in either the torsional reduction or the final recovery step. $\square$

4. SEPARATION OF VARIABLES

In this section, we prove Theorem 1.2 by exploiting the separated dependence of the shear modulus on the transverse and axial variables. Boundary determination on the bottom and lateral faces provides complementary information about this dependence. For both multiplicative and additive separation, these boundary values determine μ throughout the domain, without prior knowledge of the individual factors or summands. Once μ is known, the full elastic boundary map determines λ . We treat the two representations separately in Sections 4.1 and 4.2, and complete the recovery of both parameters in Section 4.3.

Let (λ_j, μ_j) , $j = 1, 2$, be the two coefficient pairs in Theorem 1.2, and suppose that their localized maps on $\Gamma = \Gamma_0 \cup \Gamma_\Sigma$ agree. We first derive the boundary identities used in both cases. For $W = \Gamma_0$ or $W = \Gamma_\Sigma$, the trace spaces defined in Section 2.1 satisfy

$$\mathcal{X}_W(\partial\Omega)^3 \subset \mathcal{X}_\Gamma(\partial\Omega)^3.$$

Indeed, $C_c^\infty(W) \subset C_c^\infty(\Gamma)$, and both closures are taken in $H^{1/2}(\partial\Omega)$. By the definition of the localized maps, for all $f, g \in \mathcal{X}_W(\partial\Omega)^3$,

$$\langle (\Lambda_{\lambda_1, \mu_1}^W - \Lambda_{\lambda_2, \mu_2}^W) f, g \rangle = \langle (\Lambda_{\lambda_1, \mu_1}^\Gamma - \Lambda_{\lambda_2, \mu_2}^\Gamma) f, g \rangle = 0.$$

Thus the localized maps agree separately on the bottom face and on the observed lateral face.

Both Γ_0 and Γ_Σ are nonempty relatively open subsets of the smooth part of $\partial\Omega$. The coefficients are smooth in a common neighborhood of $\overline{\Omega}$ and satisfy (1.1). Lemma 2.1 therefore applies with either choice of W . Taking $\eta = 0$ in that lemma gives

$$\begin{aligned}\mu_1(y, 0) &= \mu_2(y, 0), \quad y \in G, \\ \mu_1(y, t) &= \mu_2(y, t), \quad y \in \Sigma, \quad 0 < t < L.\end{aligned}\tag{4.1}$$

The first identity covers every transverse position in G . The second provides the axial dependence at any fixed $y_* \in \Sigma$. We use these identities to compare the unknown factors or summands.

4.1. Multiplicative separation. We prove the shear modulus assertion in case (i) of Theorem 1.2. The bottom trace determines the transverse factor up to a positive constant. The lateral trace then fixes the reciprocal scaling of the axial factor.

Proof of Theorem 1.2(i). Write $\mu_j(y, t) = \alpha_j(y)\beta_j(t)$ as in Theorem 1.2(i). Substituting these representations into the first identity in (4.1) gives

$$\alpha_1(y)\beta_1(0) = \alpha_2(y)\beta_2(0), \quad y \in G.$$

The numbers $\beta_j(0)$ are positive, so we may set

$$c = \frac{\beta_1(0)}{\beta_2(0)} > 0.$$

Dividing the bottom identity by $\beta_2(0)$ gives $\alpha_2(y) = c\alpha_1(y)$ for every $y \in G$. Since both factors are continuous on $\overline{G}$, this relation extends to

$$\alpha_2 = c\alpha_1 \quad \text{on } \overline{G}. \quad (4.2)$$

In particular, it holds on the observed arc $\Sigma \subset \partial G$.

Choose $y_* \in \Sigma$, which is possible because Σ is nonempty. The second identity in (4.1), together with (4.2), yields

$$\alpha_1(y_*)\beta_1(t) = \alpha_2(y_*)\beta_2(t) = c\alpha_1(y_*)\beta_2(t), \quad 0 < t < L.$$

Positivity of α_1 on $\overline{G}$ allows us to cancel $\alpha_1(y_*)$. We obtain $\beta_2(t) = c^{-1}\beta_1(t)$ on $(0, L)$. Continuity at both endpoints then gives

$$\beta_2 = c^{-1}\beta_1 \quad \text{on } [0, L]. \quad (4.3)$$

Combining (4.2) and (4.3), we conclude that

$$\mu_2(y, t) = (c\alpha_1(y))(c^{-1}\beta_1(t)) = \mu_1(y, t), \quad (y, t) \in \overline{G} \times [0, L]. \quad (4.4)$$

This proves recovery of μ from the localized data in case (i).

Equations (4.2) and (4.3) also prove the multiplicative assertion in Remark 1.1. Every admissible representation differs by this constant rescaling, and the rescaling preserves μ . The normalization $\beta_j(0) = 1$ fixes $c = 1$ and hence identifies the two factors separately. No normalization is needed for (4.4). $\square$

4.2. Additive separation. We next prove the shear modulus assertion in case (ii) of Theorem 1.2. The bottom trace determines the transverse summand up to an additive constant. The lateral trace forces the opposite constant in the axial summand.

Proof of Theorem 1.2(ii). Write $\mu_j(y, t) = a_j(y) + b_j(t)$ as in Theorem 1.2(ii). The first identity in (4.1) becomes

$$a_1(y) + b_1(0) = a_2(y) + b_2(0), \quad y \in G.$$

Set

$$c = b_1(0) - b_2(0).$$

Rearranging the bottom identity gives $a_2(y) - a_1(y) = c$ on G . The right-hand side is independent of y . Continuity of a_1 and a_2 therefore yields

$$a_2(y) - a_1(y) = c, \quad y \in \overline{G}. \quad (4.5)$$

Fix $y_* \in \Sigma$. The second identity in (4.1) reads

$$a_1(y_*) + b_1(t) = a_2(y_*) + b_2(t), \quad 0 < t < L.$$

Using (4.5) at the boundary point y_* , we obtain

$$b_2(t) - b_1(t) = a_1(y_*) - a_2(y_*) = -c, \quad 0 < t < L.$$

Continuity of the axial summands extends this equality to

$$b_2 = b_1 - c \quad \text{on } [0, L]. \quad (4.6)$$

It follows from (4.5) and (4.6) that

$$\mu_2(y, t) = (a_1(y) + c) + (b_1(t) - c) = \mu_1(y, t), \quad (y, t) \in \overline{G} \times [0, L]. \quad (4.7)$$

This proves recovery of μ from the localized data in case (ii).

The relations $a_2 = a_1 + c$ and $b_2 = b_1 - c$ give the additive assertion in Remark 1.1. These shifts preserve μ , and the normalization $b_j(0) = 0$ forces $c = 0$. The argument uses only differences of the summands, so it requires no separate positivity of a_j or b_j . $\square$

4.3. Recovery of the first Lamé parameter. Sections 4.1 and 4.2 establish the conclusion for partial boundary data in Theorem 1.2. It remains to prove simultaneous recovery from the full elastic map. This step is common to both separation classes.

Completion of the proof of Theorem 1.2. Suppose that

$$\Lambda_{\lambda_1, \mu_1}^\Omega = \Lambda_{\lambda_2, \mu_2}^\Omega.$$

Restricting the full maps to the trace space in Section 2.1 gives

$$\langle (\Lambda_{\lambda_1, \mu_1}^\Gamma - \Lambda_{\lambda_2, \mu_2}^\Gamma) f, g \rangle = \langle (\Lambda_{\lambda_1, \mu_1}^\Omega - \Lambda_{\lambda_2, \mu_2}^\Omega) f, g \rangle = 0$$

for all $f, g \in \mathcal{X}_\Gamma(\partial\Omega)^3$. The localized maps therefore agree. In case (i), Section 4.1 gives (4.4). In case (ii), Section 4.2 gives (4.7). Thus, in either case,

$$\mu_1 = \mu_2 \quad \text{on } \overline{\Omega}.$$

We may now apply Proposition 2.3. Indeed, $\Omega = G \times (0, L)$ has the product geometry allowed in that proposition. Both coefficient pairs are smooth in a common neighborhood of $\overline{\Omega}$ and satisfy (1.1) by the hypotheses of Theorem 1.2. Their full maps agree by assumption, and their shear moduli agree by the preceding argument. The treatment of the edges is contained in Lemma 2.2, used in the proof of Proposition 2.3. That lemma extends the two pairs to a ball while preserving equality of the shear moduli and of the full boundary maps. Proposition 2.3 consequently gives

$$\lambda_1 = \lambda_2 \quad \text{on } \overline{\Omega}.$$

Together with the equality of the shear moduli, this proves

$$(\lambda_1, \mu_1) = (\lambda_2, \mu_2) \quad \text{on } \overline{\Omega}.$$

The separation assumptions were used only to identify μ , so no such condition on λ_j enters this last step. The proof of Theorem 1.2 is complete. $\square$

5. DIRECTIONAL QUASIANALYTICITY

In this section, we prove Theorem 1.3 by combining boundary determination with quasianalyticity along a fixed direction. The boundary map determines all derivatives of μ at the boundary, and quasianalyticity allows these derivatives at an endpoint to determine μ throughout the corresponding connected line component. Applying this argument to each component gives uniqueness of μ throughout the domain without requiring convexity. The recovery of λ then follows from uniqueness for a known shear modulus. Section 5.1 proves Theorem 1.3, and Section 5.2 extends the argument to prescribed smooth curves and localized data.

5.1. Quasianalyticity along lines.

Proof of Theorem 1.3. Assume that the full elastic Dirichlet-to-Neumann maps agree. Let $h = \mu_1 - \mu_2$. Equality of the full maps implies equality of their localized restrictions on every open boundary patch. The domain is smooth, and both coefficient pairs are smooth in the common neighborhood U and satisfy (1.1). Lemma 2.1 therefore applies at every boundary point and gives

$$D^\eta h(p) = 0 \quad (p \in \partial\Omega, \eta \in \mathbb{N}_0^3). \quad (5.1)$$

Fix an arbitrary point $x \in \Omega$. Write $x = y + s_0 e$, where $y = x - (x \cdot e)e \in e^\perp$ and $s_0 = x \cdot e$. The set I_y is open in $\mathbb{R}$. Let (a, b) be its connected component containing s_0 . Since Ω is bounded, both endpoints are finite.

The point $p_a = y + a e$ lies in $\overline{\Omega}$. If it belonged to Ω , openness would contradict the maximality of the component (a, b) ; hence $p_a \in \partial\Omega$.

Choose the interval J and the common class $C^{\{M\}}(J)$ provided by the theorem for this component. Set

$$h_y(s) = \mu_1(y + s e) - \mu_2(y + s e), \quad s \in J.$$

For any compact $K \Subset J$, choose the constants $C_j, R_j > 0$ in Definition 1.1 for the two functions $s \mapsto \mu_j(y + s e)$. The common weight M gives

$$\sup_{s \in K} |h_y^{(k)}(s)| \leq (C_1 + C_2) \max\{R_1, R_2\}^k k! M_k, \quad k \geq 0.$$

Since K was arbitrary, $h_y \in C^{\{M\}}(J)$. For each integer $k \geq 0$, the chain rule gives

$$h_y^{(k)}(a) = (e \cdot \nabla)^k h(p_a) = \sum_{|\eta|=k} \frac{k!}{\eta!} e^\eta D^\eta h(p_a) = 0. \quad (5.2)$$

Here $\eta! = \eta_1! \eta_2! \eta_3!$ and $e^\eta = e_1^{\eta_1} e_2^{\eta_2} e_3^{\eta_3}$. The last equality follows from (5.1) at $p_a \in \partial\Omega$, and for $k = 0$ the expression is simply $h(p_a)$. Since $h \in C^\infty(U)$, its boundary derivatives are the restrictions of its ambient derivatives. The point a is an interior point of J because J is open and contains $[a, b]$. Applying the quasianalytic identity principle (1.6) to (5.2) gives $h_y = 0$ throughout J , and in particular on (a, b) . Hence $h(x) = h_y(s_0) = 0$.

Since x was arbitrary, $\mu_1 = \mu_2$ in Ω , and continuity gives equality on $\overline{\Omega}$. The argument applies to each connected line component separately. It uses neither convexity of Ω nor uniform bounds for the quasianalytic classes on different lines. The use of ambient directional derivatives also covers tangential endpoints.

Set $\mu = \mu_1 = \mu_2$ in Ω . The original full data equality now reads

$$\Lambda_{\lambda_1, \mu}^\Omega = \Lambda_{\lambda_2, \mu}^\Omega.$$

The original pairs are smooth in U and strongly convex on Ω . Since Ω is bounded, connected, and smooth, Proposition 2.3 gives $\lambda_1 = \lambda_2$ on $\overline{\Omega}$. Thus $(\lambda_1, \mu_1) = (\lambda_2, \mu_2)$ on $\overline{\Omega}$, with quasianalyticity used only for the shear modulus. $\square$

5.2. Quasianalyticity along curves. As noted in Remark 1.3, the argument of Section 5.1 extends to prescribed smooth curves and localized boundary data.

Proposition 5.1. *Let Ω , U , and (λ_j, μ_j) , $j = 1, 2$, satisfy the domain, smoothness, and strong convexity conditions of Theorem 1.3. Let $\Gamma \subset \partial\Omega$ be nonempty and relatively open, possibly $\Gamma = \partial\Omega$. Suppose that Ω is covered by the images $\gamma((a, b))$ of a family of smooth curves $\gamma : J \rightarrow U$, where $a < b$, $J \supset [a, b]$ is a connected open interval, and*

$$\gamma(a) \in \Gamma, \quad \gamma((a, b)) \subset \Omega, \quad \gamma(b) \in \overline{\Omega}, \quad \gamma'(s) \neq 0 \quad (a \leq s \leq b).$$

For each curve, suppose that

$$\mu_j \circ \gamma \in C^{\{M\}}(J), \quad j = 1, 2,$$

for a common class as in Definition 1.1. The curves, their parametrizations, and the weights are prescribed independently of the candidates. The intervals and weights may vary between curves, and the defining constants need not be uniform. Then

$$\Lambda_{\lambda_1, \mu_1}^\Gamma = \Lambda_{\lambda_2, \mu_2}^\Gamma \implies \mu_1 = \mu_2 \quad \text{on } \overline{\Omega}.$$

If the full elastic Dirichlet-to-Neumann maps agree, then both Lamé parameters agree on $\overline{\Omega}$. No quasianalyticity is required of λ_j .

Proof. Suppose that the localized maps agree and set $h = \mu_1 - \mu_2$. Lemma 2.1 gives

$$D^\eta h(p) = 0 \quad (p \in \Gamma, \eta \in \mathbb{N}_0^3). \quad (5.3)$$

Fix $x \in \Omega$ and choose a covering curve γ and $s_0 \in (a, b)$ such that $x = \gamma(s_0)$. Set $p = \gamma(a) \in \Gamma$ and $q = h \circ \gamma$. By hypothesis, the two compositions $\mu_j \circ \gamma$ belong to the same class $C^{\{M\}}(J)$. The estimate for differences in Section 5.1 therefore gives $q \in C^{\{M\}}(J)$. Smoothness of h and γ suffices for the following chain rule calculation. Starting from $q'(s) = \nabla h(\gamma(s)) \cdot \gamma'(s)$, repeated differentiation gives

$$q^{(k)}(a) = \sum_{1 \leq |\eta| \leq k} A_{k, \eta}(a) D^\eta h(p) = 0, \quad k \geq 1, \quad (5.4)$$

where each coefficient $A_{k, \eta}$ is a polynomial in derivatives of γ of order at most k . Equation (5.3) makes every factor $D^\eta h(p)$ zero and also gives $q(a) = h(p) = 0$. Together with (5.4), this shows that all derivatives of q vanish at a . Since a is an interior point of the connected interval J , the quasianalytic identity principle (1.6) gives $q = 0$ on J . In particular, $h(x) = q(s_0) = 0$. As x was arbitrary, continuity yields $\mu_1 = \mu_2$ on $\overline{\Omega}$.

Suppose now that the full elastic maps agree. By the definition in Section 2.1, for all $f, g \in \mathcal{X}_\Gamma(\partial\Omega)^3$,

$$\langle (\Lambda_{\lambda_1, \mu_1}^\Gamma - \Lambda_{\lambda_2, \mu_2}^\Gamma) f, g \rangle = \langle (\Lambda_{\lambda_1, \mu_1}^\Omega - \Lambda_{\lambda_2, \mu_2}^\Omega) f, g \rangle = 0.$$

Thus the localized maps agree, and the preceding argument yields $\mu_1 = \mu_2$ on $\overline{\Omega}$. The domain is bounded, connected, and smooth. Both coefficient pairs are smooth in the common neighborhood U and satisfy (1.1). Together with equality of the full maps and the common shear modulus, these conditions allow us to apply Proposition 2.3, giving

$$\lambda_1 = \lambda_2 \quad \text{on } \overline{\Omega}.$$

Hence both Lamé parameters agree on $\overline{\Omega}$, as asserted in Proposition 5.1. $\square$

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